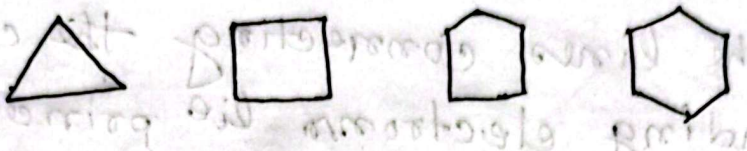
17.06.25

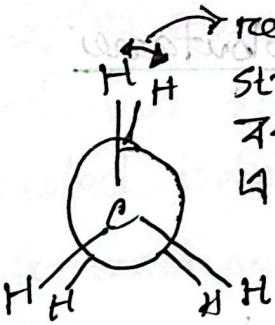
CH-2 Configuration and conformation of cyclic Compounds.

SUZON-50

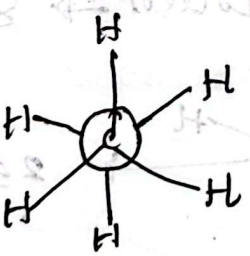


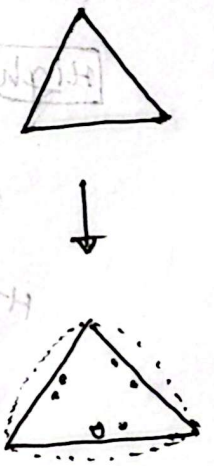
Conformation of Cyclopropane

Cyclopropane suffers from both angle & torsional strain. Each C–C bond is eclipsed due to rigidity of the molecule. Three bon angle (60°) sp^3 hybridized.



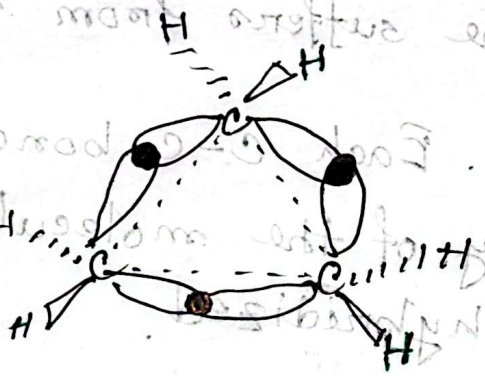
repulsion হওয়ার কারনে strain হয় এবং C–C bond বরাবর rotate করে standard form এ যায়।





হালকা bent হবে। তখন e⁻=e⁻ distance বড় হবে।

The highest e⁻ density of the C-C bonds doesn't lie along the lines connecting the carbon atoms. Bonding electrons lie principally outside of the triangular internuclear lines resulting in bend bonds.

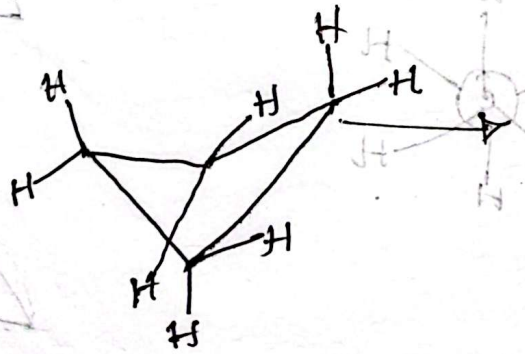


Conformation of Cyclobutane



Angle strain → 107 - 90°

High torsional strain → 8 H_s



25° Above the plane.

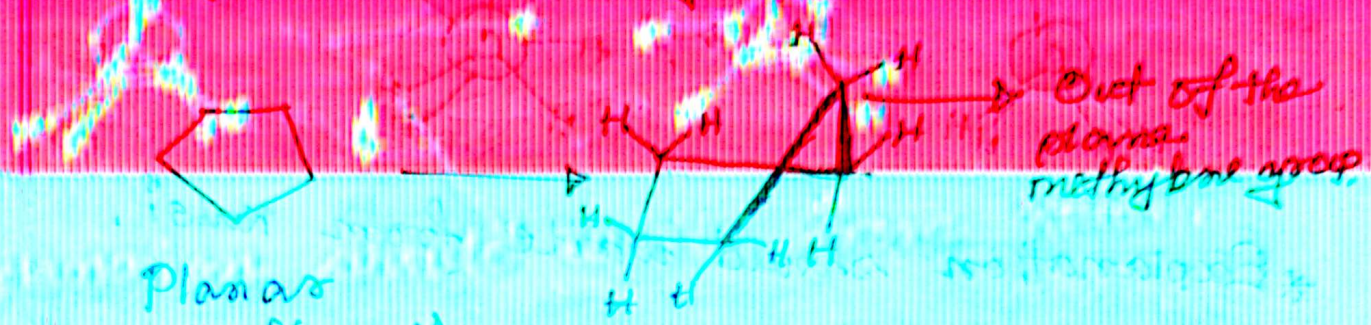
Folded structure.



Explanation should come from Kabi

The extra stability of the puckered structure and non-bonded interaction of the oxygens of angle in deformation (increased angle strain) Raman spectra and electron diffraction experiment confirm the puckered conformation.

Conformation of Cyclopentane



Planar (Eclipsed)

Envelop/Boat shape

Angle strain 108°

Torsional → 10 Hs highest.

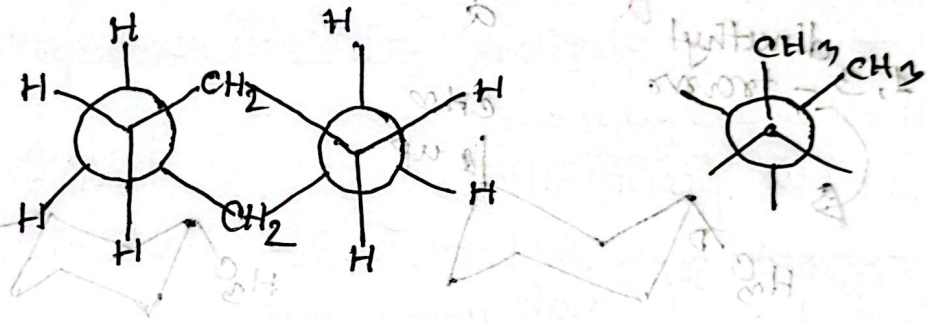
Explanation: If one considers a planar pentagonal structure of cyclopentane, one would have C-C-C bond angles 108°, which is so close to 109°30", However, all the Hs are completely eclipsed and it would have about 10 kcal/mol of strain energy.

So, the molecule finds distort from a planar conformation even though it increases the angle strain. The actual structure is of "envelope" shape. The out-of-plane methylene group is approximately staggered with respect to its neighbours. By twisting about the various C-C bonds, successive conformations are realised in which four carbons are in a plane and 5th is out of plane.

Conformation of Cyclohexane:

① Cyclohexane molecule exists predominantly in non-planar, pucked conformation, the chair conformation.

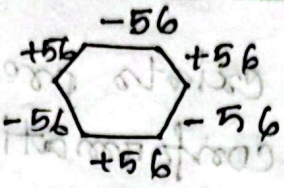
② All bonds are staggered, therefore pitzer strain is minimized. However, a gauche butane like interactions betwn neighbouring methylene groups lead to steric strain.

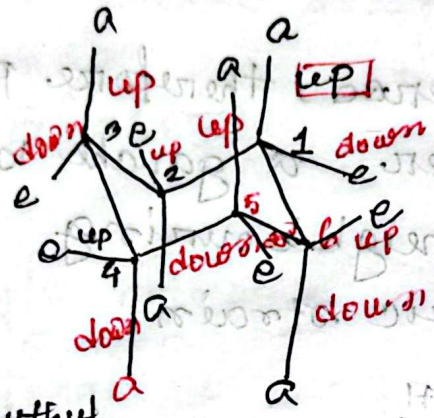


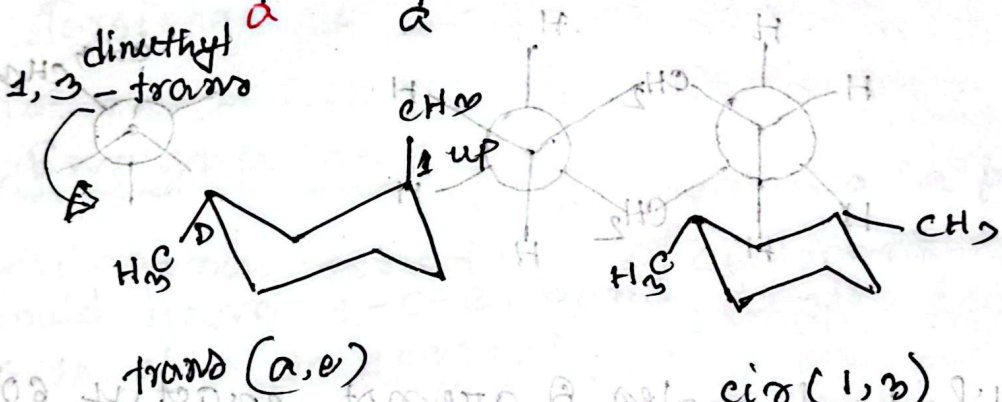
③ The dihedral angles θ are not exactly 60°, but these are around 56°

④ Each ring C-C bond is labelled with a numbe corresponding to dihedral angle and the sig corresponding to clockwise (+) or anticlockwis (-) for the angle.

⑤ The bond angles are not exactly 109.3 but 111°.

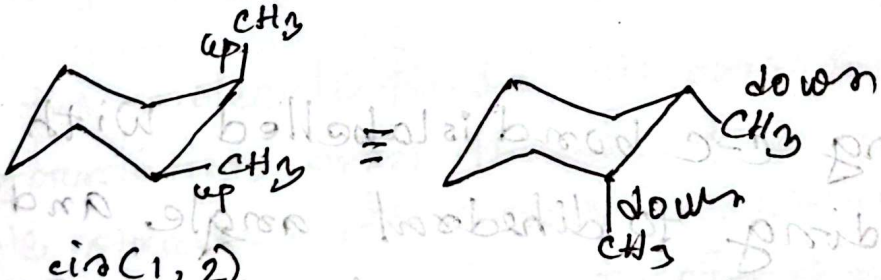






trans (a,e)

cis (1,3)

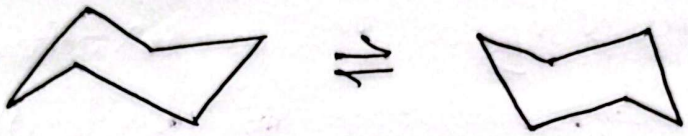


cis (1, 2)

cis (1, 2)

cis (1, 4)

Ring flip: মানে হলো – 1 ও আগানো অথবা পিছানো।



Conformational Analysis of Cyclohexane

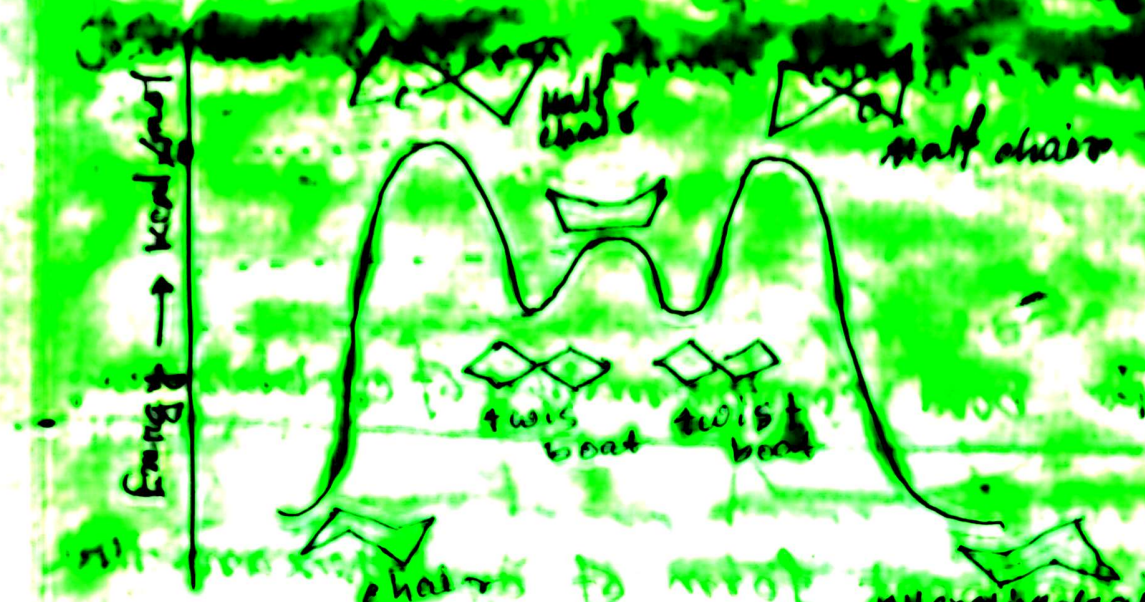


Fig: The A E for cyclohexane in ring conversion.

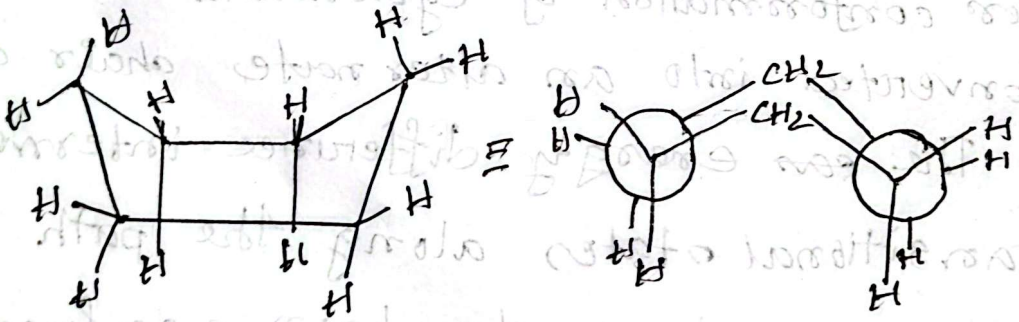
One chair conformation of cyclohexane can be easily converted into an alternate chair conformation. The energy difference intermediates and transitional states along the path of one chair to the alternate chair conformation provides another example of conformational analysis.

The two chairs and twisted boat

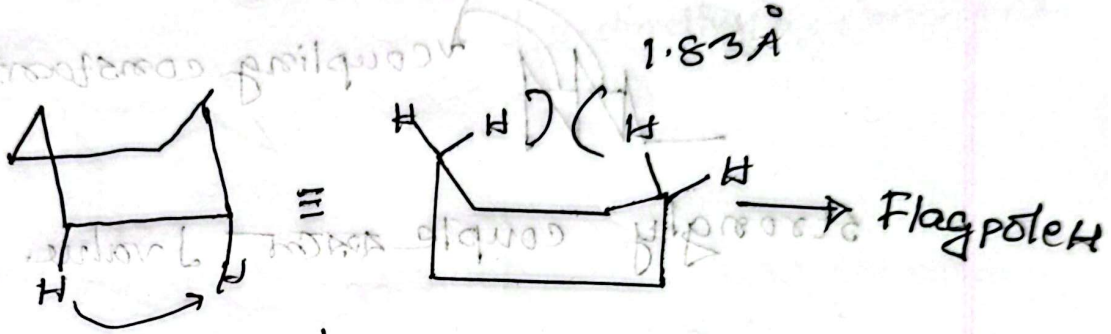
conformations are inter mediates represents energy minima and the two half chairs and full boat conformation represents the transitional state (energy maxima).

The boat conformation of cyclohexane:

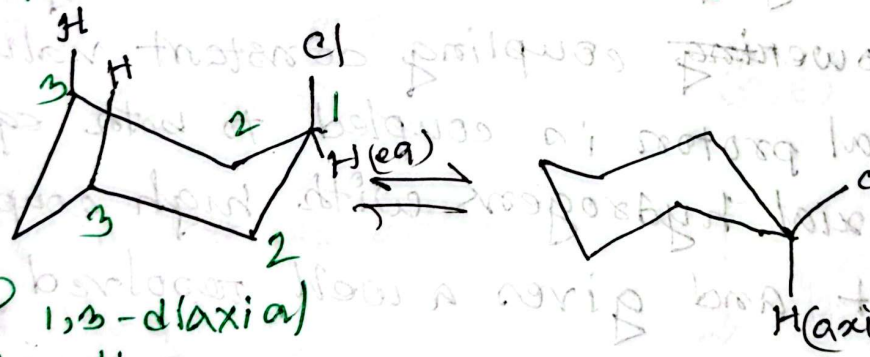
① In the boat form of cyclohexane there is complete eclipsing of the hydrogens attached to the carbon atoms forming side CH_2-CH_2 bonds.



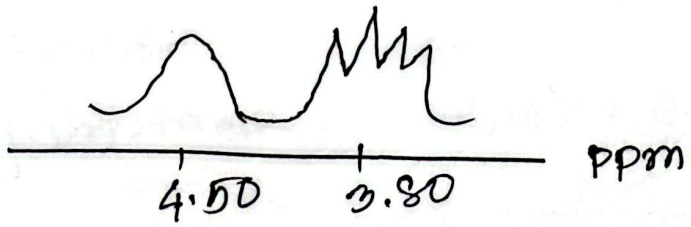
② The flagpole hydrogens interfere with each other sterically and there is a 2nd factor which leads to instability of the boat form in comparison to chair. The flagpole hydrogens are only 1.83Å apart.



Conformation of chlorocyclohexane at -15



2(b) 1,3-d(axial) interaction.



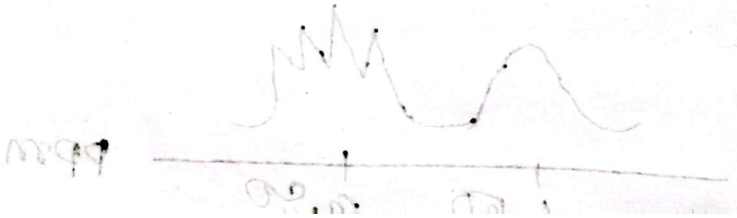
@ Pavia
* Robiul sir

equitorial H এর সাথে H গুলোকে sense করে
সামান্য ধরা broad peak হয়। (Left)
Right – conformation এ উল্টে গেলে এদের
multiplicity হয়।

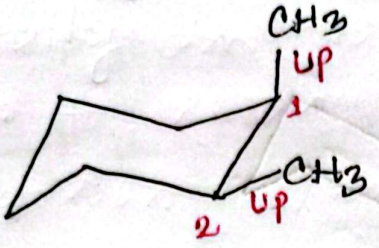


strongly couple করলে J value বেশি হয়।

The equatorial proton is coupled with neighbouring protons which are all gauche with lowering coupling constant value. The axial proton is coupled to both equatorial / axial hydrogens with high coupling constant and gives a well resolved multiplets.



Conformation of disubstituted cyclohexane

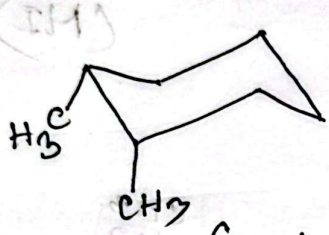


cis (a,e)

cis হলে up–up or down–down.

1,2-dimethyl cyclohexane

or

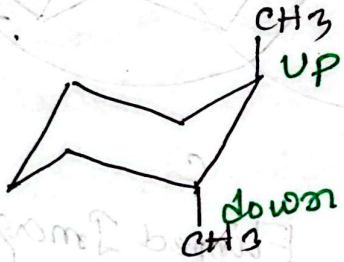


(e,a)

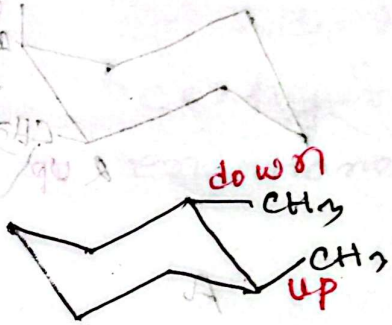
trans :

up–down

down–up

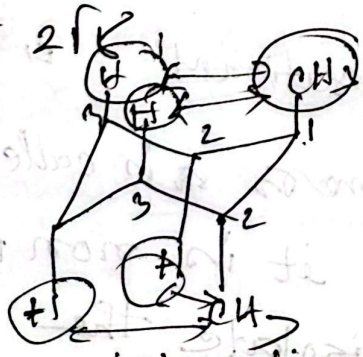


(a,a)



(e,e)

1,3 diaxial interaction



interaction

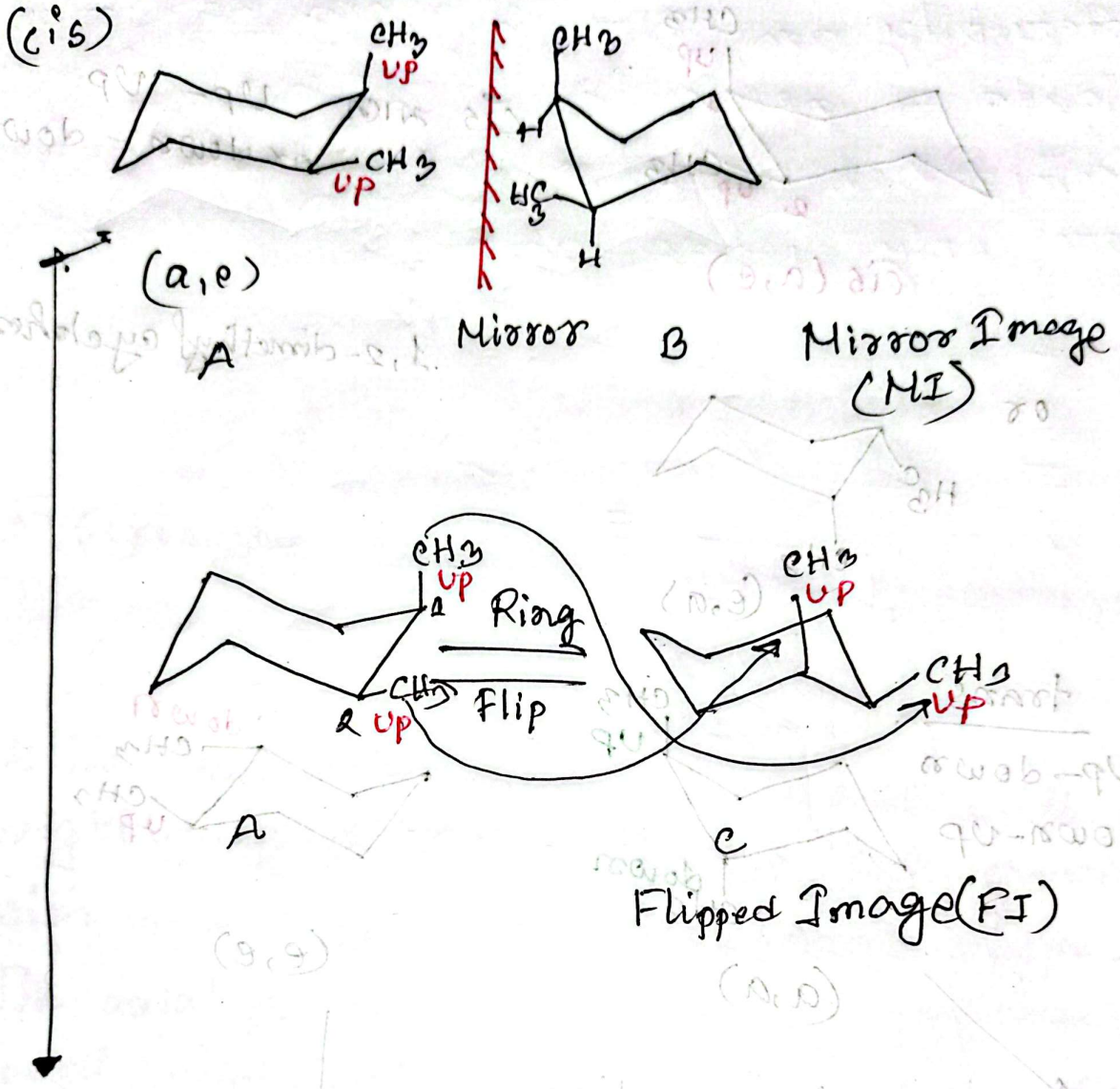
very unstable.

Most stable. There is no steric hindrance in e,e position.

So,

(e,e) > (a,e) ≈ (e,a) > (a,a)

Optical Isomerism:



MI & FI same কেন conformational enantiomer.

~~Enantio~~

" " " FI " configurational "

Mixture of 1:1 enantiomers কে is called Racemic mixture and it is non resolvable i.e. we can not isolate the enantiomers. If temperature is so lower than it can be isolated.

Different

A → a → e Up – Up clockwise

B → a → e Up – Up Anti clockwise

C → a → e Up – Up Anti clockwise

B = C

↓

A ≠ B

Different & non-superimposable mirror image. So they are enantiomer.

Conformational Enantiomers

Individually optically active.

1:1 reaction. Racemic mixture

Optically inactive. And they are non-Resolvable.

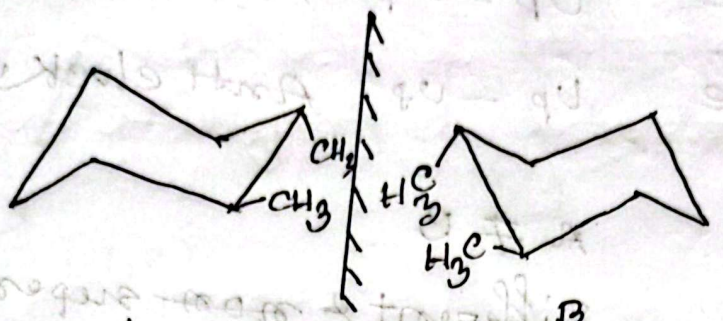
A ≠ C Non configurational enantiomers.

cis : Up – Up / down – down

trans: Up – down / down – Up

[A & B are superimposable by chair flipping and therefore, constitute an optically inactive dl-pair unresolvable because they interconvert very rapidly.]

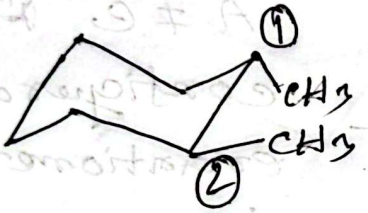
Trans (1,2)



A
(e,e)

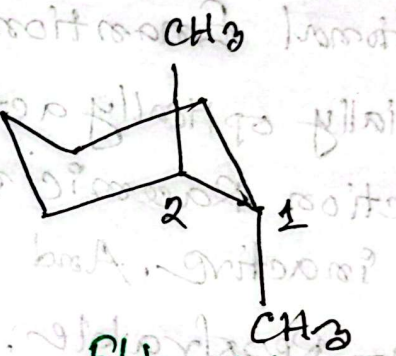
B
(e,e) (MI)

Enantiomer



A

Flip
ring



Flipped image
(C) (FI)

A → (e,e) → D, up → Clockwise

B → (e,e) → D, up → Anticlockwise

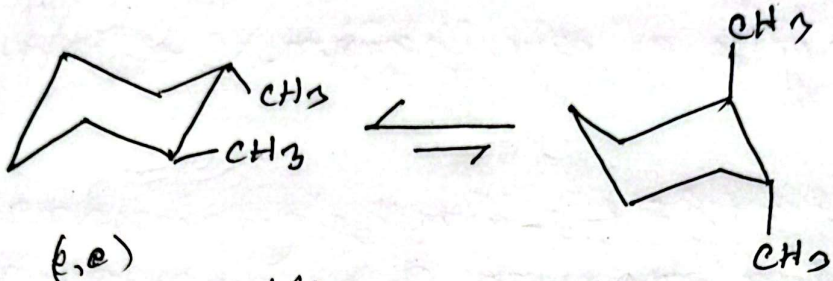
C → (a,a) → D, up → clockwise

A ≠ B (Different)

pair of enantiomer.

B ≠ C configurational Enantiomer.

Resolvable enantiomer.



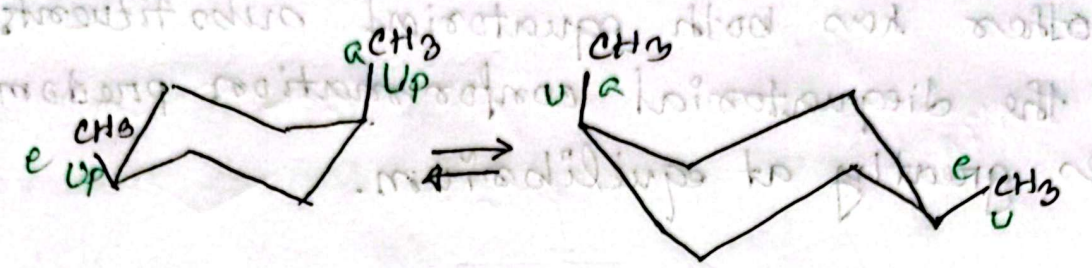
(e,e)
More stable.

বেশির পরিমাণে থাকে
তাই ⟵ (large)
এরও বেশিরভাগ থাকে।

(a,a) less stable.
4 fr interaction হয়।
যার কারণে very unstable.
& কম পরিমাণে থাকে।
তাই- Room Temp. এ খুব
optically active (যেসমস্ত
কম্পাউন্ড)

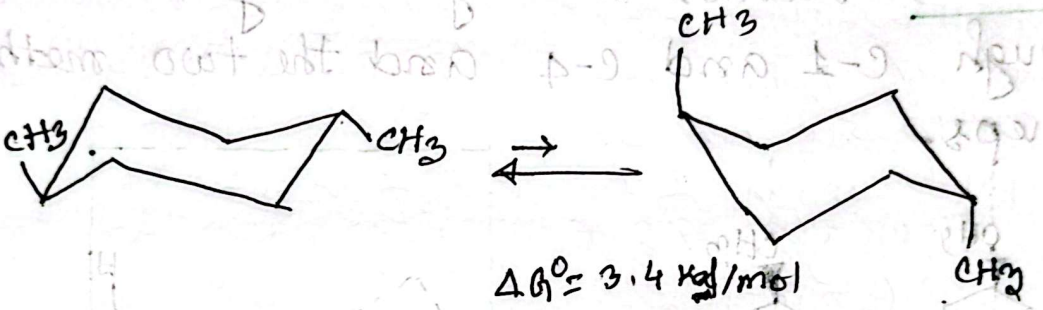
The trans isomer is usually more stable when the substituents are both equatorial e.g. for X = CH_3, The trans isomer is more stable than the cis isomer, in which one substituent is necessarily in the unfavourable axial position; e.g. for X = CH
ΔG is about 1.8 Kcal/mol

Cis 1,4- dimethylcyclohexane



$\Delta G^0 = 0$

trans-1,4- dimethylcyclohexane.



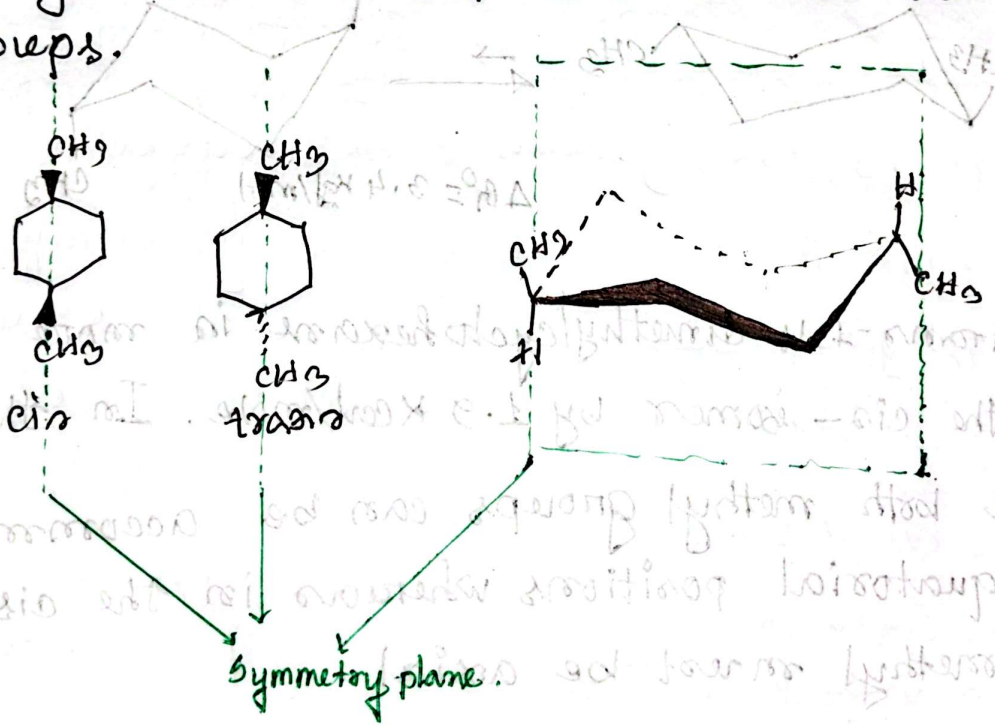
$\Delta G^0 = 3.4$ Kcal/mol

The trans-1,4-dimethylcyclohexane is more stable than the cis- isomer by 1.9 Kcal/mole. In the trans isomer both methyl groups can be accommodated in equatorial positions whereas in the cis isomer one methyl must be axial.

Cis and trans-1,4- dimethyl cyclo hexane can exist in two chair conformations. In the case of cis. isomer, the two conformations are of equal energy, since each has one axial and one equatorial substituent.

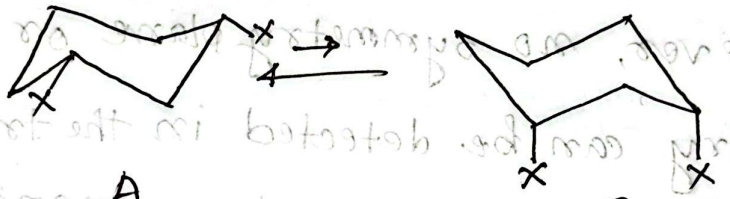
For. trans, one has both axial while the other has both equatorial substituents and the diequatorial conformation predominates greatly at equilibrium.

Cis and trans-1,4-dimethylcyclohexane as a pair of diastereomers. Both compounds are achiral, because the symmetry plane passes through C-1 and C-4 and the two methyl groups.

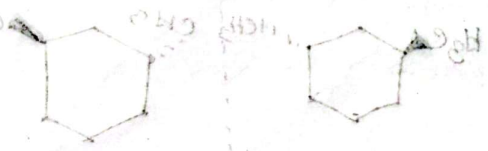


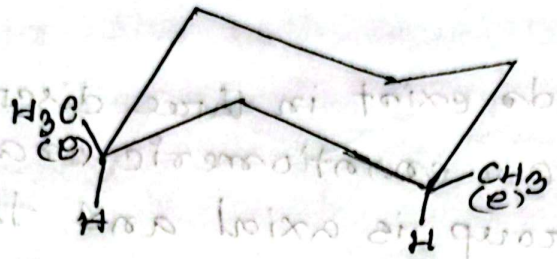
1,3- disubstituted cyclohexane

The 1,3- compounds exist in three discrete stereo-isomeric forms, an enantiomeric trans pair in which one group is axial and the other equatorial, and a cis isomer which has a plane of symmetry and has interconvertible conformational isomers having both groups respectively A & B and A predominates.

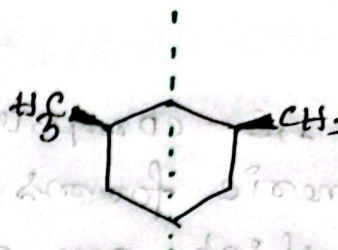


In the case of 1,3- dimethylcyclohexane a plane of symmetry is easily detectable in the cis- isomer. Since this compound (in contrast to the 1,4-dimethyl isomer) has chiral conformers, cis-1,3- dimethylcyclohexane is, therefore, a meso compound.



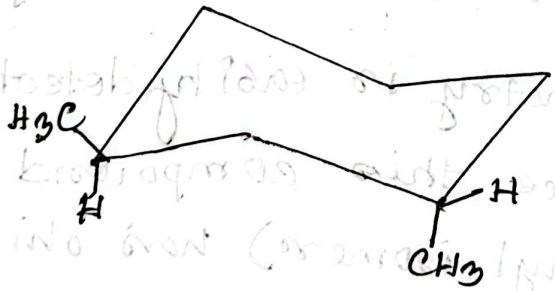


cis-1,3-dimethyl cyclohexane
both methyls equatorial.
(More stable)



Plane of symmetry
(Meso compound)

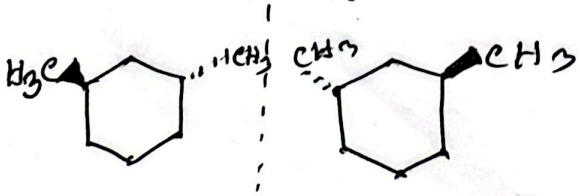
However, no symmetry plane or center of symmetry can be detected in the trans isomer. Thus, there are two enantiomeric trans-1,3-dimethylcyclohexanes.



[Less stable than cis isomer.]

trans-1,3-dimethylcyclohexane
one methyl equatorial

Mirror



A pair of enantiomers.